RUIJI SUN

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EDUCATION

PhD **University of California at Berkeley, Building Science and Technology** 2026
Thesis:
Advisor: Gail S. Brager, Stefano Schiavon

MS **Carnegie Mellon University, Sustainable Design** 2020
Thesis: "BIM and BEM Based Sustainable Architecture Design"
Advisor: Dana Cupkova, Omer T. Karaguzel

BE **Xi'an University of Arch & Tech, Mechanical Engineering (HVAC)** 2017
Thesis: "Modeling Experiment of Air Environment in Metro Station"
Advisor: Angui Li

BA **Xi'an University of Arch & Tech, Architecture Design** 2017
Design: "Art Museum Design"
Advisor: Li An

HONORS AND AWARDS

1st Prize ETAMSF Scholarship, ASHRAE GOLDEN GATE 2020
Eric T Andresen Memorial Scholarship Fund

Research Project Fellowship, NEMA 2019
Developed **BIM content** guidelines for electrical products

Research Project Fellowship, ASHRAE 2019
Improved the interoperability of gbXML between BIM to BEM

Merit Award, SOLAR DECATHLON CHINA 2017
Participated as Xi'an University of Arch & Tech team member

Architecture Travel Grant, CAPOL 2016
Studied advanced pre-fabrication building technology

1st Prize in Competition, China Refrigeration & Air-Conditioning Industry 2016
Designed innovative LNG bus cooling systems

1st Prize in Competition, LianZhu Beijing Technology Inc. 2015
Designed and reinvented the news stand architecture

3rd Prize National tournament of green building innovation, CIOB 2015
Design green building utilizing recyclable materials and raining water

Innovation Scholarship, Xi'an University of Arch & Tech 2015
Performed excellently in science and technology

RESEARCH EXPERIENCE

Center for The Built Environment, Berkeley, CA 2019
Graduate Research Scientist, Advisor: Paul Raftery

- Developed software for Mortar data analytics
- Evaluated long-term thermal comfort by continuous Mortar data
- Enabled reproducible data analysis in building science.

ASHRAE, Pittsburgh, PA 2020
Data Scientist, Advisor: Weili Xu

- Researched data mapping rules of IDF and gbXML schema
- Programmed in Revit-Dynamo environment to convert HVAC systems data
- Programmed in BuildSim API to streamline cloud building energy simulation
- Developed 10 building geometry test cases to improve BIM software
- Interviewed 20 BIM and BEM industry professionals to collect issues

NEMA, Pittsburgh, PA 2019
Data Scientist, Advisor: Krishnan Gowri

- Developed industry guidance to standardize electrical products data in BIM
- Researched different data structures to improve BIM models

Center for Building Performance & Diagnostics, Pittsburgh, PA 2019
Student, Advisor: Omer Karaguzel

- Co-simulated building energy consumption and control logics in MATLAB
- Developed and design an online dashboard with real-time data using Django
- Visualized environmental data both spatially and temporally using Matplotlib
- Measured environmental data using National Environmental Assessment Toolkit cart
- Evaluated occupants' thermal comfort by Post-Occupancy Evaluation (POE) survey

Tongji University, Shanghai, China 2017
Student, Advisor: Philip F. YUAN

- Researched VR and AR technology for CFD visualization
- Researched different architecture formation by wind visualization
- Programmed a mechanical model in wind tunnel using C++

Xi'an University of Arch & Tech, Xi'an, China 2017
Student, Advisor: Augui Li

- Analyzed piston wind airflow patterns in metro station
- Designed an experiment platform to model air environment in metro station
- Measured airflow velocity in different metro station

TEACHING EXPERIENCE

Carnegie Mellon University, Pittsburgh, PA 2018
Teaching Assistant, School of Architecture

- Instructed students to analyze HVAC systems and calculate energy consumption based on data from the OSIsoft PI database.
- Assisted professor Volker Hartkopf contacting and communicating with building science industries in China.

PUBLICATIONS

Sun, R. J., Hu, Z. Z., Xu, W. L., & Gowri, K. 2020, "Improving the interoperability of gbXML through redefining data mapping rules of HVAC systems," *ASHRAE 2020 Annual Conference*.

Hu, Z. Z., Sun, R. J., Xu, W. L., & Gowri, K. 2020, "Obstacles that gbXML is facing: analysis on BIM software interoperability survey," *ASHRAE 2020 Annual Conference*.

PROFESSIONAL TRAINING

LEED AP BD+C
Leadership in Energy and Environmental Design Accredited Professional

SKILL SET

Data Analytics: SQL, Pandas, NumPy

Programming: Python, HTML&CSS, MATLAB

Simulation: EnergyPlus, OpenStudio, DesignBuilder, Fluent

Modeling: AutoCAD, Revit, Rhino, Grasshopper

Design: Hand Drawing, Adobe Suite